2.6 Polynomial Functions

So far we have understood about linear functions and quadratic functions. Now we shall generalize these ideas. We call an expression of the form $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ is called a ***polynomial*** in the variable x, where $a_i \in \mathbb{R}$, $i = 0, 1, 2, \cdots, n$. Here n is a non-negative integer. When $a_n \neq 0$, we say that the polynomial has **degree** n. The numbers $a_0, a_1, \ldots, a_n \in \mathbb{R}$ are called the coefficients of the polynomial. The number a_0 is called the constant term and a_n is called the leading coefficient (when it is non-zero). It is clear that:

(i) $100x^7 - \pi x^5 + 20\sqrt{2}x^2 + 7x + 1.22$ is a polynomial of degree 7.
(ii) $(17x - 3)(x + 3)(2x - \sqrt{\pi})(x + 2.3)$ is a polynomial of degree 4.
(iii) $(x^2 + x + 1)(x^3 + 2x + 2)(x^5 - 5x + \sqrt{3})$ is a polynomial of degree 10.

One may substitute specific values for x, say $x = c$ and obtain $a_n c^n + a_{n-1}c^{n-1} + \cdots + a_1 c + a_0$. A function of the form $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ is called a ***polynomial function*** which is defined from $\mathbb{R}$ to $\mathbb{R}$. We shall treat polynomial and polynomial function as one and the same.

A polynomial with degree 1 is called a ***linear polynomial***. A polynomial with degree 2 is called a ***quadratic polynomial***. A **cubic polynomial** is one that has degree three. Likewise, degree 4 and degree 5 polynomials are called ***quartic*** and ***quintic*** polynomials respectively. Note that any constant $a \neq 0$ is a polynomial of degree zero!

Two polynomials $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$, $a_n \neq 0$ and $g(x) = b_m x^m + b_{m-1}x^{m-1} + \cdots + b_0$, $b_m \neq 0$ are equal if and only if $f(x) = g(x)$ for all $x \in \mathbb{R}$. It can be proved that $f(x) = g(x)$ if and only if $n = m$ and $a_k = b_k, k = 0, 1, 2, \cdots n$. Given two polynomials, one can form their **sum** and **product**. For example if $P(x) = 2x^3 + 7x^2 - 5$ and $Q(x) = x^4 - 2x^3 + x^2 + x + 1$, then $P(x) + Q(x) = x^4 + 8x^2 + x - 4$ (by adding the corresponding coefficients of the like powers of x) and $P(x)Q(x) = 2x^7 + 3x^6 - 12x^5 + 4x^4 + 19x^3 + 2x^2 - 5x - 5$ by multiplying each term of $P(x)$ by every term of $Q(x)$. It is easy to see that the degree of $P(x)Q(x)$ is the sum of the degrees of $P(x)$ and $Q(x)$, whereas the degree of $P(x) + Q(x)$ is at most the maximum of degrees of $P(x)$ and $Q(x)$. Here is an example of the graph of a cubic polynomial function.

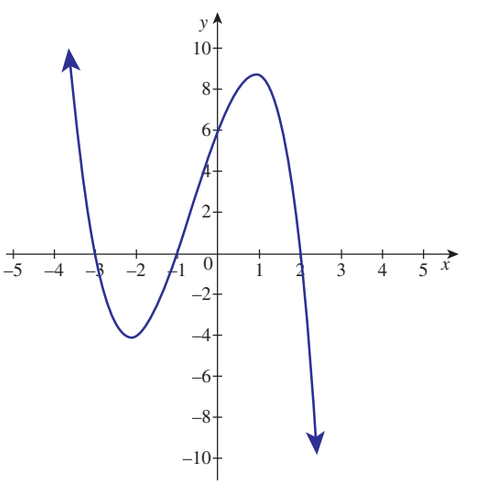


Figure 2.6

Suppose that $f(x)$ and $g(x)$ are polynomials where $g(x)$ is not zero. The quotient $\frac{f(x)}{g(x)}$ is called a **rational function,** which is defined for all $x \in \mathbb{R}$ such that $g(x) \neq 0$. In general, a rational function need not be a polynomial.